

LLM-judge rubric for feature abstractness

Purpose. The triviality metrics (modal-word %, distinct-word count) only measure *lexical* diversity — they can't tell a noise feature that fires on many varied words apart from a genuine *abstract* feature like "negation." This LLM judge measures **abstractness/complexity** directly, so we can compare the abstractness *distribution* across the three SAEs (full / hybrid / outbias).

How it's run.

- The judge is an LLM (currently `gpt-5.6-terra`, via the Batch API) rating one feature at a time.
- It is **blind to which SAE a feature came from** — so the rubric cannot bias the full-vs-hybrid-vs-outbias comparison.
- For each feature it sees two blocks of examples: **PEAK** (strongest activations) and **TYPICAL** (a uniform random sample of firings), and rates them independently. Each example shows the activating token wrapped in `⟨⟩` with a few tokens of context.
- Output per block: two LLM-rated axes (coherence, abstractness) + a short `label` and `rationale`. **Breadth** is *not* LLM-rated — it's measured in Python (see below).

How to read the examples

- **Judge from the `⟨⟩` token itself** — ask "what property of *this* token makes the feature fire?" The surrounding words are only context. Rate what the highlighted tokens share, not what the paragraphs happen to be about.
- **Subwords.** The `⟨⟩` token is often just a *piece* of a word (the tokenizer splits words, e.g. `Cr|umble`, `Vie|ja`). Read the whole word and the small construction it completes, not the bare fragment — the shared property may live there (e.g. `um|ble` / `Dip` / `Soup` / `Cr` all being "the final piece of a dish name in a recipe title" → coherent). But do **not** infer a property the token or its word doesn't itself carry.

Rate COHERENCE first — it gates abstractness

COHERENCE (1–5)

Do the `⟨⟩` tokens in this block share one consistent property (a meaning, a role, or a surface form) that explains the firing?

1	none: highlighted tokens look unrelated; no property explains them (noise)
2	weak: a minority share a property; most do not
3	mixed: a clear property holds for most, with several outliers
4	strong: nearly all firings share one clear property
5	exact: every firing has the same clear property, no exceptions

ABSTRACTNESS (0–5)

Given a coherent property, how abstract is it? Rate the property named for coherence.

0	No real pattern. Use whenever coherence ≤ 2 — don't invent a concept out of noise
1	The exact same word or spelling every time — fires on one specific token or letter-pattern, regardless of meaning (e.g. always <code>the</code> ; or always words ending in <code>-ing</code>)
2	One word or its close variants — a single word across its senses, or a few synonyms for one thing (e.g. big / large / huge)
3	One topic — many different words, but all from the same subject area (e.g. healthcare, elections, basketball)
4	A role or relationship, not a topic — many unrelated words linked by what they <i>do</i> in the sentence, not what they're <i>about</i> (e.g. negation words, comparisons, or "an organization being founded," which shows up in sports, business, and charities alike)
5	An abstract idea no word list could capture — spans many topics and roles at once (e.g. uncertainty, formality, politeness)

3-vs-4 test. Look at what the passages are *about*. If firings stay locked to one subject (sports words never appear outside sports) → **3**. If the same relation/action/role recurs across many *different* subjects → **4**. A noun you could name a subject ("about sports/law") is a topic (3); a verb-y action or relation that happens in any subject ("something is being founded / negated / compared") is a frame (4).

Breadth — measured in Python, not by the LLM

Breadth (how many distinct words a feature fires on) is the **effective-#words** metric — `exp(entropy)` of the feature's word distribution — computed over *all* firings in `eval_trivial.py`, not rated by the LLM. A deterministic count over the full data is exact and cheaper than having the model eyeball it from a handful of examples, so the LLM is reserved for the two axes that genuinely need judgment (coherence + abstractness).

Then give

- **label** — 2–6 words naming what the feature detects (write "incoherent" if coherence ≤ 2).
 - **rationale** — one sentence naming the concrete shared property of the 《》 tokens (or stating there is none), citing the actual property, not a vibe.
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Lineage (Scaling Monosemanticity)

This reuses the paper's automated-interpretability recipe — an LLM judge on a small anchored scale, grounded in the activating token, with a "no interpretation here" floor (0), validated against human labels — but points it at **abstractness** rather than the paper's **specificity**. The abstractness ladder and the coherence-gates-abstractness structure are specific to this project (they're the axis word-counting can't reach).

Validation status

Validated against hand labels on a seeded, SAE-balanced sample of features (blind, same rubric). Agreement was strong on coherence, breadth, and the "no-pattern" gate; the weakest axis was abstractness, traced to two causes: (a) the judge under-crediting features that are semantically coherent but lexically varied, and (b) subword tokenization. The **subword rule above** was added to address these. A fresh-sample re-validation (a held-out seed, so the fix isn't measured on the features it was tuned on) is the next step before scaling to the full feature set.